

Spaces:

Notation:

$\mathbb{R} \rightarrow$ The Real Numbers / Scalars, The complete ordered field.

$\mathbb{R}^2 \xrightarrow{\text{Cartesian Plane}} \rightarrow$ Set of all Pairs of real numbers

$$= \{ (x, y) \mid x, y \in \mathbb{R} \}$$

$$\mathbb{R}^3 = \{ (x, y, z) \mid x, y, z \in \mathbb{R} \}$$

$$\mathbb{R}^n = \{ (x_1, \dots, x_n) \mid x_i \in \mathbb{R} \}$$

Definition *

Let $P = (p_1, \dots, p_n)$ & $Q = (q_1, \dots, q_n)$ be two points in $\mathbb{R}^n$. i.e. distance

$$d(P, Q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2} \rightarrow d = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

Pythagorean theorem

Vectors

* A vector is a directed line segment in $\mathbb{R}^n$ modulo location.
 ↳ line in space has a head & a tail
 ↳ if direction of line doesn't change they are the same vector
 ↳ you can add up vector forces in physics if they are the same

* vector notation

$$v = \vec{v} = \langle v_1, v_2, \dots, v_n \rangle = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \rightarrow \text{good for vector equations}$$

v_i 's are the components of v , the displacements in each dimension.

vector operations:

* Magnitude turns vector $\rightarrow$ scalar
 $|v| = \sqrt{\sum_{i=1}^n v_i^2}$

The mag or length of a vector is a nonnegative scalar.

It is non-degenerate which just means

$$v = 0 \iff |v| = 0$$

* Scaling:

(vector) $\rightarrow$ preserves direction & changes length

Scalar multiplication: as really scalar-vector multiplication, simply scales each component by a common factor

$$c\vec{v} = \langle cv_1, cv_2, \dots, cv_n \rangle$$

$$|c\vec{v}| = |c| \cdot |v|$$

↳ have to have a positive or 0 result.

* Normalization: $\rightarrow$ means if unit vector $= 1$
 A unit vector has magnitude 1.

$$|u| = 1$$

* Every non zero vector $v \neq 0$ can be normalized by dividing by its magnitude.

$$u = \frac{v}{|v|} = \frac{1}{|v|} \cdot v$$

You can divide a vector by a scalar as long as it's not 0.

multiply by reciprocal of its magnitude

* Addition

vector addition: work shows you think it would

$$v + w = \langle v_1 + w_1, \dots, v_n + w_n \rangle$$

* A sum of scaled vectors is called a linear combination. is the fundamental vector expression.

$$av + bw$$

$$\sum_{i=1}^n c_i v_i$$

I stopped listening didn't understand

←